

Mathematics A

Unit 1 • Chapter OL-T58 Classified

Cross-session • chapter-organised candidate

26 classified items • 95 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 1 • Unit 1 • Chapter OL-T58 • 26 items • 95 marks

Chapter	Items	Marks	Starts
01. OL-T58 Combined & Conditional Probability	26	95	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Combined & Conditional Probability

Unit 1 • 95 marks • 26 item(s)

June 2025 | 4MA1-1H Q21 | 3 marks

Calculate an exact composition in three or more draws

June 2025 | 4MA1-2H Q20 | 6 marks

Calculate a direct given-that probability

June 2024 | 4MA1-1H Q13 | 5 marks

Find exactly one, same or different independent outcomes

June 2024 | 4MA1-1H Q16 | 4 marks

Use a complement in repeated no-replacement selection

November 2023 | 4MA1-1H Q20 | 4 marks

Find exactly one, same or different independent outcomes

November 2023 | 4MA1-2H Q11 | 5 marks

Find exactly one, same or different independent outcomes

June 2023 | 4MA1-2H Q13 | 4 marks

Find the probability that all independent events occur

June 2023 | 4MA1-2H Q20 | 3 marks

Use a complement in repeated no-replacement selection

June 2022 | 4MA1-1H Q12 | 4 marks

Use branch-dependent conditional probabilities

June 2022 | 4MA1-1H Q15 | 4 marks

Find exactly one, same or different independent outcomes

June 2022 | 4MA1-2H Q16 | 8 marks

Calculate a direct given-that probability

January 2022 | 4MA1-1H Q14 | 2 marks

Find the probability that all independent events occur

January 2020 | 4MA1-1H Q18 | 4 marks

Use a complement in repeated no-replacement selection

January 2020 | 4MA1-2H Q14 | 2 marks

Calculate an unordered two-selection event

January 2020 | 4MA1-2H Q18 | 2 marks

Use a histogram or table as a conditional sample space

... 11 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Working Unit 14

4MA1-1H Q21

ORIGINAL QUESTION**3 marks**

21 A box contains 20 counters.

9 of the counters are red

7 of the counters are yellow

4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

.....
(Total for Question 21 is 3 marks)

Worked Solution 14

Exactly two counters of the same colour

UNIT 1
3 marks**Use the complement**

$$P(\text{exactly two}) = 1 - P(\text{all same}) - P(\text{all different}).$$

All same

$$P(\text{all same}) = \frac{9 \cdot 8 \cdot 7 + 7 \cdot 6 \cdot 5 + 4 \cdot 3 \cdot 2}{20 \cdot 19 \cdot 18} = \frac{738}{6840}.$$

All different

$$P(\text{all different}) = \frac{6(9 \cdot 7 \cdot 4)}{20 \cdot 19 \cdot 18} = \frac{1512}{6840}.$$

Finish

$$1 - \frac{738 + 1512}{6840} = \frac{4590}{6840} = \frac{51}{76}.$$

Final answer

$$\frac{51}{76}$$

Dependency check Sampling is without replacement; the factor 6 counts the orders of three different colours.

Working Unit 23

4MA1-2H Q20

ORIGINAL QUESTION**6 marks**

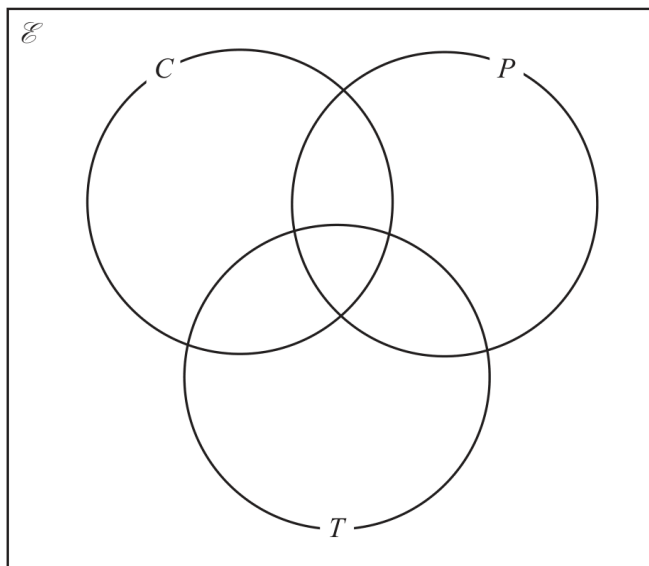
- 20** 120 gardeners were asked if they grow carrots (C) or potatoes (P) or tomatoes (T)

Of these gardeners

- 43 grow carrots
- 12 grow carrots and potatoes and tomatoes
- 18 grow carrots and potatoes
- 27 grow carrots and tomatoes
- 32 grow potatoes and tomatoes
- 29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

- (a) Complete the Venn diagram to show this information.



(3)

- (b) Find $n(T' \cap C)$

(1)

One of the gardeners who grows carrots is chosen at random.

- (c) Calculate the probability that this gardener also grows potatoes.

(2)

Worked Solution 23

Three-set Venn diagram and conditional probability

UNIT 1
6 marks**Intersections**

$$C \cap P \cap T = 12, CP \text{ only} = 6, CT \text{ only} = 15, PT \text{ only} = 20.$$

Single-set regions

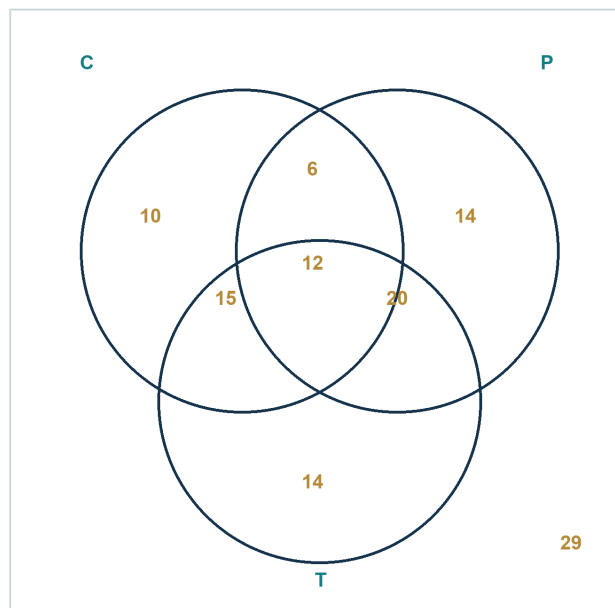
$$C \text{ only} = 43 - (6 + 15 + 12) = 10. \text{ If } P \text{ only} = T \text{ only} = k, 10 + 6 + 15 + 12 + 20 + 2k = 120 - 29, \text{ so } k = 14.$$

Part (b)

$$n(T' \cap C) = 10 + 6 = 16.$$

Part (c)

$$P(P \mid C) = \frac{n(P \cap C)}{n(C)} = \frac{18}{43}.$$



Solved reference diagram

Final answer

Regions (10, 14, 14, 6, 15, 20, 12; outside 29), 16, $\frac{18}{43}$

Dependency check Parts (b) and (c) read the completed diagram; the diagram appears once.

4MA1-1H Q13

ORIGINAL QUESTION

5 marks

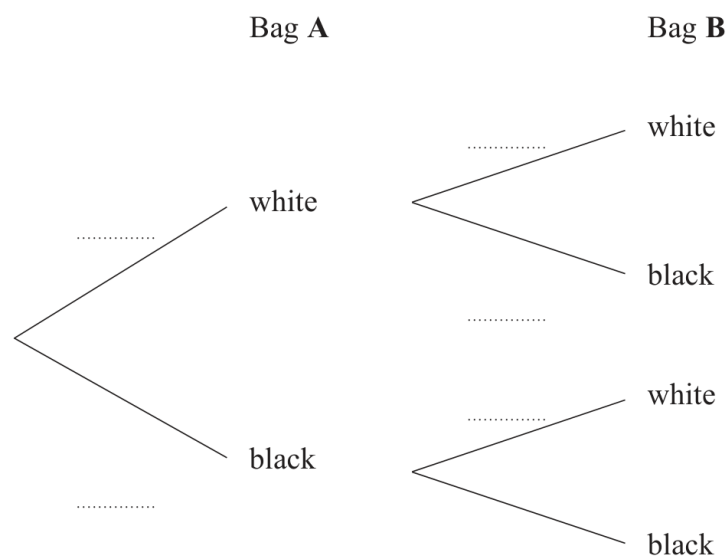
13 Harman has two bags of beads.

In bag **A**, there are 3 white beads and 7 black beads.

In bag **B**, there are 5 white beads and 4 black beads.

Harman takes at random a bead from bag **A** and a bead from bag **B**

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Harman takes two beads of the same colour.

(3)

Worked Solution 10

Probability tree

MODULAR UNIT 1

5 marks

Branches

$$A : (3/10, 7/10), \quad B : (5/9, 4/9).$$

Same colour

$$(3/10)(5/9) + (7/10)(4/9) = 43/90.$$

Final answer

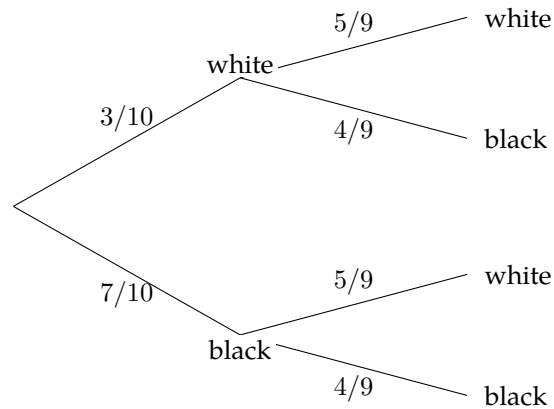
$$\frac{43}{90}$$

Completed diagram 10

Probability tree

MODULAR UNIT 1

5 marks

Completed solution diagram: Probability tree

Question 11

4MA1-1H Q16

ORIGINAL QUESTION

4 marks

16 There are 20 sweets in a box.

15 of the sweets are red

5 of the sweets are yellow

Fred takes at random 3 sweets from the box.

Work out the probability that Fred takes at least one sweet of each colour from the box.

Worked Solution 11

At least one of each colour

MODULAR UNIT 1

4 marks

Complement

$$1 - \frac{15}{20} \frac{14}{19} \frac{13}{18} - \frac{5}{20} \frac{4}{19} \frac{3}{18} = \frac{45}{76}.$$

Final answer

$$\frac{45}{76}$$

Question 14

4MA1-1H Q20 demand

ORIGINAL QUESTION UNIT 1

4 marks

20 A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag.
He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

.....

Worked solution 14

Probability of mean 15 cents

MODULAR UNIT 1

4 marks

Demand – Probability of mean 15 cents

Find each coin probability

$$P(10) = \sqrt{49/121} = 7/11, \quad P(20) = 4/11.$$

Use the two orders

$$P(10, 20 \text{ or } 20, 10) = 2 \times \frac{7}{11} \times \frac{4}{11} = \frac{56}{121}.$$

Final answer

56/121

Question 23

4MA1-2H Q11 (a), (b)

ORIGINAL QUESTION UNIT 1

5 marks

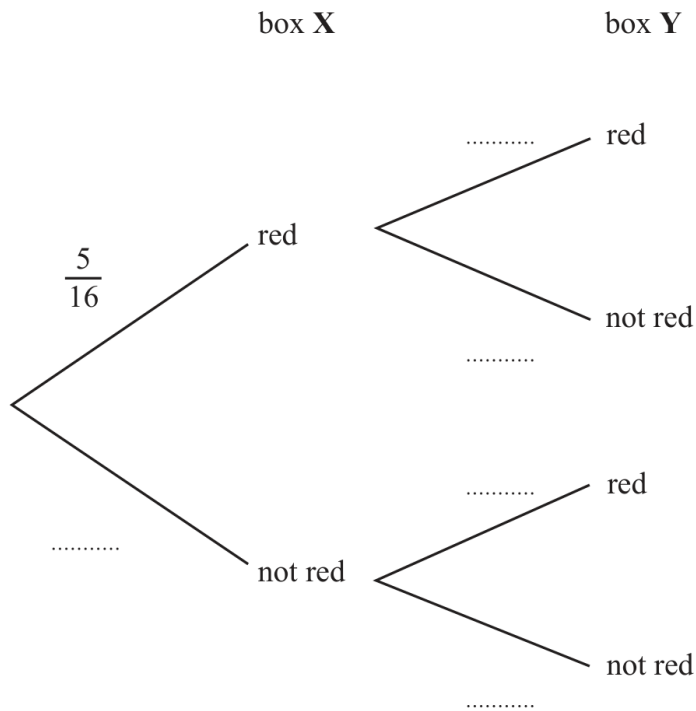
11 Sid has 2 boxes of crayons, box **X** and box **Y**

5 of the 16 crayons in box **X** are red.

7 of the 20 crayons in box **Y** are red.

Sid takes at random one crayon from box **X** and one crayon from box **Y**

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Sid takes two crayons that are red or two crayons that are not red.

(3)

Worked solution 23

Complete a two-box probability tree / Probability of two matching colours

MODULAR UNIT 1

5 marks

(a) – Complete a two-box probability tree

Complete box X

$$P(X \text{ not red}) = 1 - \frac{5}{16} = \frac{11}{16}.$$

Complete box Y on both branches

$$P(Y \text{ red}) = \frac{7}{20}, \quad P(Y \text{ not red}) = \frac{13}{20}.$$

Final answer

X not red = 11/16; each Y branch = 7/20 and 13/20

(b) – Probability of two matching colours

Add the two matching paths

$$P = \frac{5}{16} \cdot \frac{7}{20} + \frac{11}{16} \cdot \frac{13}{20}.$$

Combine

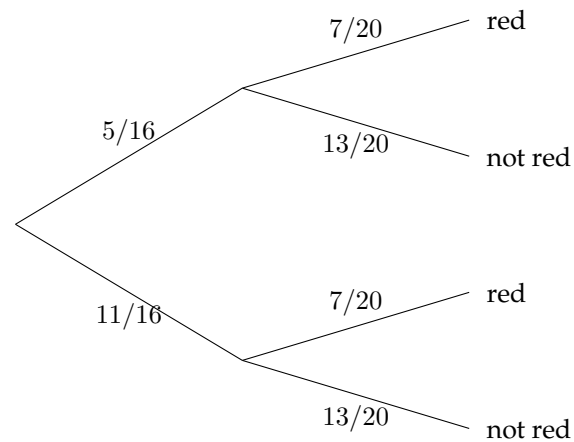
$$P = \frac{35 + 143}{320} = \frac{178}{320} = \frac{89}{160}.$$

Final answer

89/160

Completed diagram 23

Complete a two-box probability tree / Probability of two matching colours

MODULAR UNIT 1**5 marks****Completed solution diagram: Completed probability tree**

Question 18

4MA1-2H Q13

ORIGINAL QUESTION**4 marks**

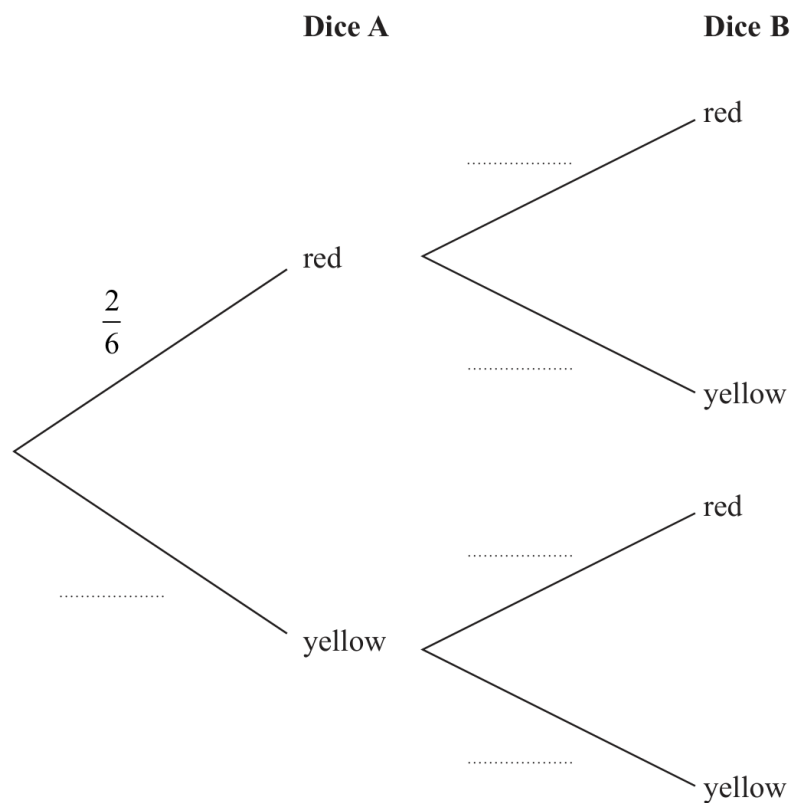
13 Narin has two fair 6-sided dice.

Dice **A** has 2 red faces and 4 yellow faces.

Dice **B** has 1 red face and 5 yellow faces.

Narin is going to throw each dice once.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that both dice land on yellow.

(2)

Worked Solution 18

Two-dice probability tree

MODULAR UNIT 1

4 marks

Complete the tree $A : R = 2/6, Y = 4/6; \quad B : R = 1/6, Y = 5/6$ on every second branch.**Both yellow**

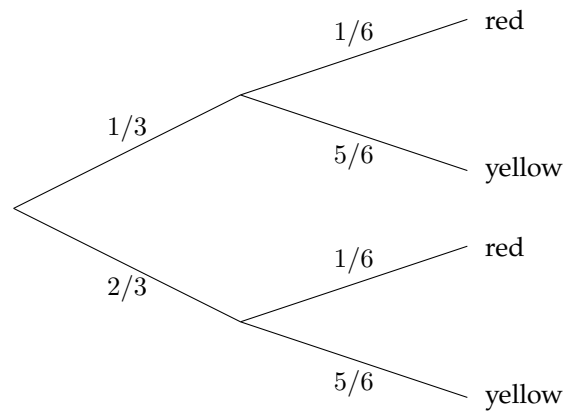
$$P(Y, Y) = \frac{4}{6} \times \frac{5}{6} = \frac{20}{36} = \frac{5}{9}.$$

Final answer

$$\frac{5}{9}$$

Completed diagram 18

Two-dice probability tree

MODULAR UNIT 1**4 marks****Completed solution diagram: Completed probability tree****Final answer**

$$P(\text{both yellow}) = \frac{2}{3} \times \frac{5}{6} = \frac{5}{9}$$

Question 25

4MA1-2H Q20

ORIGINAL QUESTION

3 marks

20 There are 12 counters in a bag.

3 of the counters are red

9 of the counters are green

Ameya, Jack and Ella each take at random one counter from the bag.

Work out the probability that at least one red counter is still in the bag.

Worked Solution 25

At least one red counter remains

MODULAR UNIT 1

3 marks

Use the complement

The only failure is drawing all three red: $P(RRR) = \frac{3}{12} \times \frac{2}{11} \times \frac{1}{10} = \frac{1}{220}$.

Subtract from one

$$P(\text{at least one red remains}) = 1 - \frac{1}{220} = \frac{219}{220}.$$

Final answer

$$\frac{219}{220}$$

Original question 09

4MA1-1H Q12 Q12(a), Q12(b)

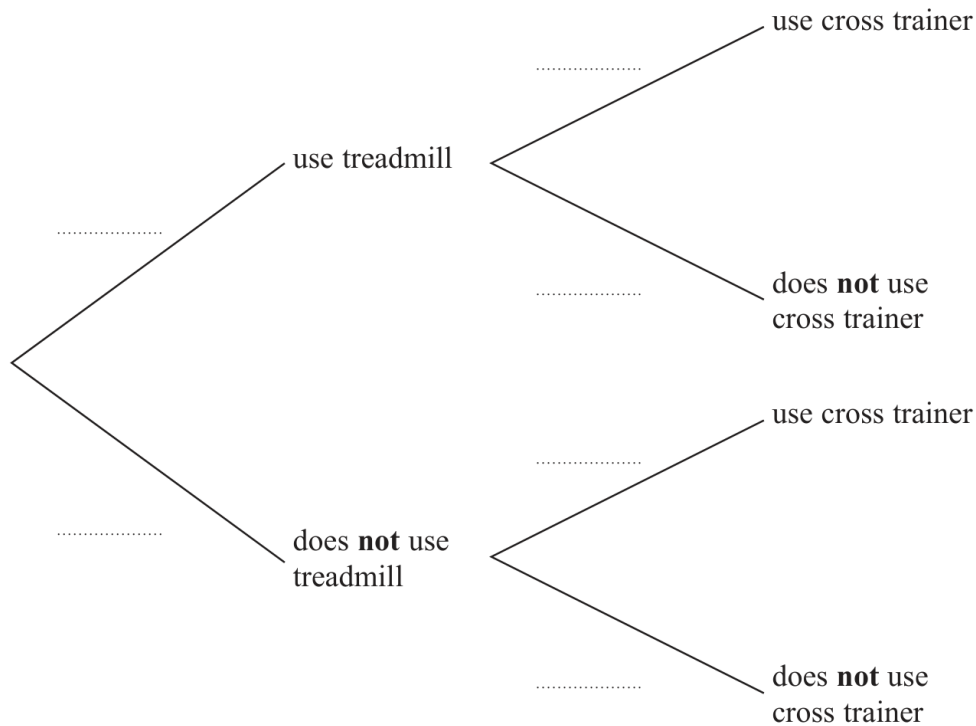
ORIGINAL QUESTION**4 marks****12** Rudolf goes to the gym.

The probability that he will use the treadmill is 0.8

When he uses the treadmill, the probability that he will use the cross trainer is 0.3

When he does **not** use the treadmill, the probability that he will use the cross trainer is 0.6

(a) Complete the probability tree diagram for this information.



(2)

(b) Work out the probability that Rudolf uses both the treadmill and the cross trainer.

.....
(2)

Worked solution 09

Chapter: Probability Diagrams - Venn & Tree Diagrams; Combined & Conditional Probability

MODULAR UNIT 1

4 marks

Q12(a) – Chapter: Probability Diagrams - Venn & Tree Diagrams

Family: Construct and read probability trees • Variant: Construct a tree from conditional wording

Method

The first split is treadmill 0.8 and no treadmill 0.2.

*Why: Complementary probabilities on a split add to 1.***Method**

After treadmill, use cross trainer 0.3 and not 0.7; after no treadmill, use cross trainer 0.6 and not 0.4.

*Why: Complete each pair with its complement.***Final answer****tree probabilities:** 0.8/0.2; then 0.3/0.7 and 0.6/0.4**Q12(b) – Chapter: Combined & Conditional Probability**

Family: Use conditional probability and restricted sample spaces • Variant: Use branch-dependent conditional probabilities

MethodThe path for both activities is treadmill followed by cross trainer, with probability 0.8×0.3 .*Why: Multiply probabilities along one complete path.***Method** $0.8 \times 0.3 = 0.24$.*Why: Evaluate the path product.***Final answer****probability:** 0.24

Completed diagram 09

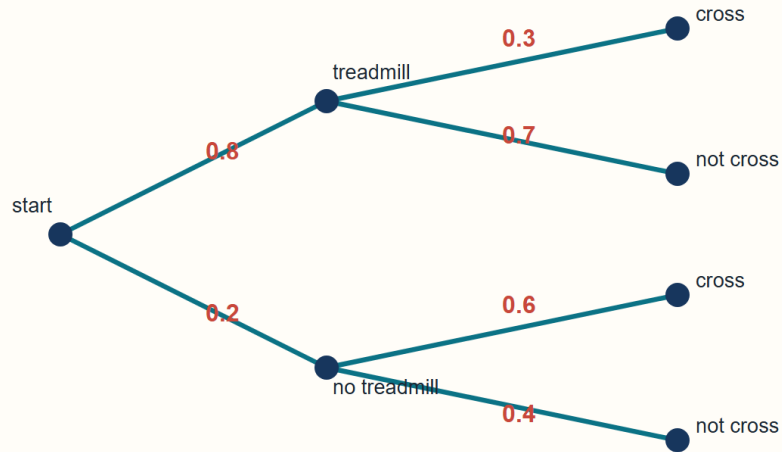
Chapter: Probability Diagrams - Venn & Tree Diagrams; Combined & Conditional Probability

MODULAR UNIT 1

4 marks

Q12(a) Completed probability tree

Place each complement beside its branch

Each pair is complementary: $0.8+0.2=1$, $0.3+0.7=1$, $0.6+0.4=1$.

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Original question 10

4MA1-1H Q15 Q15

ORIGINAL QUESTION**4 marks**

- 15** Abraham is going to play a computer game.
Abraham can win the game, draw the game or lose the game.

For any game that Abraham plays

the probability that he wins the game is 0.3
the probability that he draws the game is 0.5
the probability that he loses the game is 0.2

When Abraham wins a game, he scores +10 points.

When Abraham draws a game, he scores 0 points.

When Abraham loses a game, he scores −5 points.

Abraham plays 3 games and the points he scores in each of the 3 games are added together to get his total score.

Work out the probability that when he has played 3 games his total score is 0 points.

Worked solution 10

Chapter: Combined & Conditional Probability

MODULAR UNIT 1

4 marks

Q15 – Chapter: Combined & Conditional Probability

Family: Combine independent events • Variant: Find exactly one, same or different independent outcomes

MethodA total of zero occurs for three draws, with probability $0.5^3 = 0.125$.*Why: Three zero-point outcomes add to zero.***Method**It also occurs for one win and two losses because $10 - 5 - 5 = 0$.*Why: Identify the other score combination that totals zero.***Method**There are 3 positions for the win, so this probability is $3 \times 0.3 \times 0.2^2 = 0.036$.*Why: Count all permutations of one win and two losses.***Method**The required probability is $0.125 + 0.036 = 0.161$.*Why: The two cases are mutually exclusive, so add their probabilities.***Final answer**

probability: 0.161

Original question 24

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c)

ORIGINAL QUESTION**8 marks**

16 100 farmers are asked if they have goats (G), sheep (S) or chickens (C) on their farms.

Of these farmers

31 have sheep

53 have chickens

6 have goats, sheep and chickens

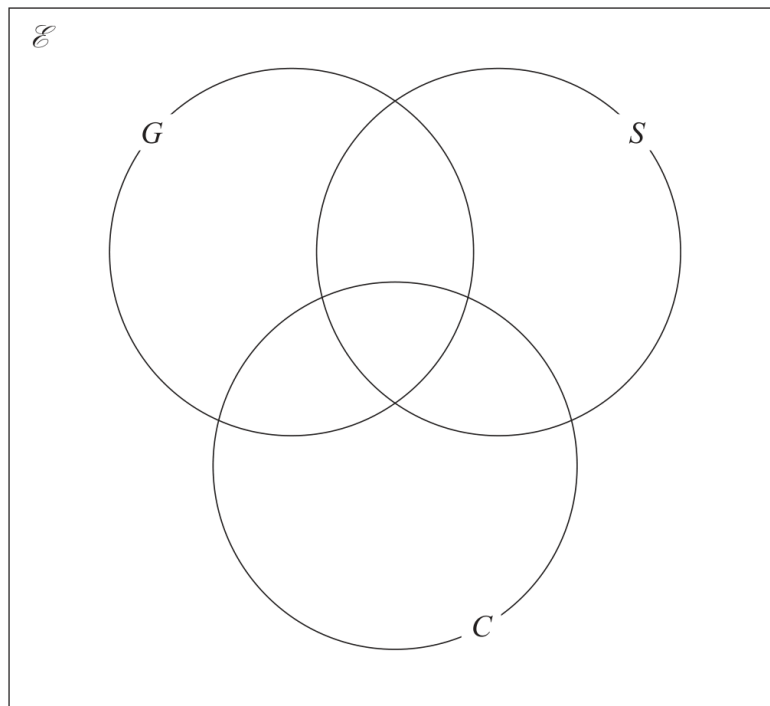
11 have sheep and goats

17 have sheep and chickens

18 have goats and chickens

20 do not have any goats, sheep or chickens

(a) Using this information, complete the Venn diagram to show the number of farmers in each appropriate subset.

**(3)**

(b) Find

Original question 24 (continued)

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c)

ORIGINAL QUESTION**8 marks**

(i) $n(G)$

(1)

(ii) $n([G \cup S]')$

(1)

(iii) $n(G' \cap C)$

(1)

One of the farmers who has chickens is chosen at random.

(c) Find the probability that this farmer also has goats.

(2)

Worked solution 24

Chapter: Probability Diagrams - Venn & Tree Diagrams; Combined & Conditional Probability

MODULAR UNIT 1

8 marks

Q16(a) – Chapter: Probability Diagrams - Venn & Tree Diagrams

Family: Represent and read events with Venn diagrams • Variant: Complete a three-set Venn diagram

Method

Put 6 in the triple intersection; then GS-only = $11 - 6 = 5$, SC-only = $17 - 6 = 11$, and GC-only = $18 - 6 = 12$.

Why: The pair totals include the triple intersection.

Method

S-only = $31 - (5 + 11 + 6) = 9$ and C-only = $53 - (11 + 12 + 6) = 24$.

Why: Subtract every already-filled sheep or chicken region.

Method

There are 80 farmers inside the three sets, so G-only = $80 - (9 + 24 + 5 + 11 + 12 + 6) = 13$; put 20 outside.

Why: Use the universal total and the 20 with none.

Final answer

Venn regions: G only 13; S only 9; C only 24; GS only 5; GC only 12; SC only 11; GSC 6; outside 20

Q16(b)(i) – Chapter: Probability Diagrams - Venn & Tree Diagrams

Family: Represent and read events with Venn diagrams • Variant: Read regions and probabilities from a Venn diagram

Method

$n(G) = 13 + 5 + 12 + 6 = 36$.

Why: Add every region inside G.

Final answer

$n(G)$: 36

Q16(b)(ii) – Chapter: Probability Diagrams - Venn & Tree Diagrams

Family: Represent and read events with Venn diagrams • Variant: Read regions and probabilities from a Venn diagram

Method

$(G \cup S)'$ contains C-only and the outside region, so $n((G \cup S)') = 24 + 20 = 44$.

Why: The complement excludes every point in G or S.

Final answer

count: 44

Q16(b)(iii) – Chapter: Probability Diagrams - Venn & Tree Diagrams

Family: Represent and read events with Venn diagrams • Variant: Read regions and probabilities from a Venn diagram

Method

G' intersection C contains C-only and SC-only, so $n(G' \text{ intersection } C) = 24 + 11 = 35$.

Why: Include chicken regions that are outside G.

Final answer

count: 35

Q16(c) – Chapter: Combined & Conditional Probability

Family: Use conditional probability and restricted sample spaces • Variant: Calculate a direct given-that probability

Method

The chosen farmer comes from the 53 farmers with chickens.

Why: This is the conditional sample space.

Method

Of these, 18 also have goats, so the probability is $18/53$.

Why: Use the G-and-C count over the C count.

Final answer

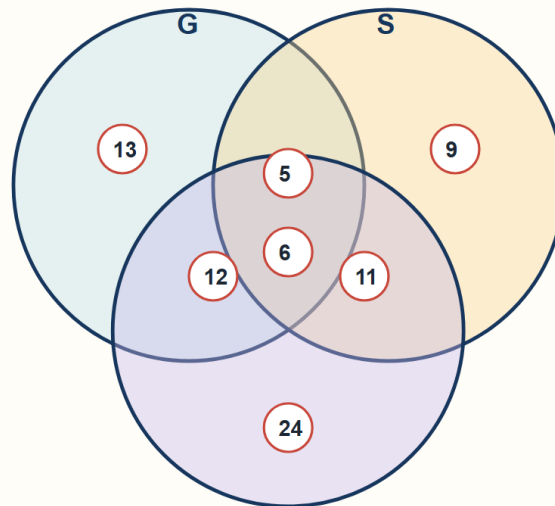
probability: $18/53$

Completed diagram 24

Chapter: Probability Diagrams - Venn & Tree Diagrams; Combined & Conditional Probability

MODULAR UNIT 1**8 marks****Q16(a) Completed three-set Venn diagram**

Place every region, including the outside count



outside

20

Inside all three = 6

Inside at least one = 80

Total farmers = 100

 $80 + 20 = 100 \checkmark$

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Original question 13

4MA1-1H Q14 Q14(b)

ORIGINAL QUESTION

2 marks

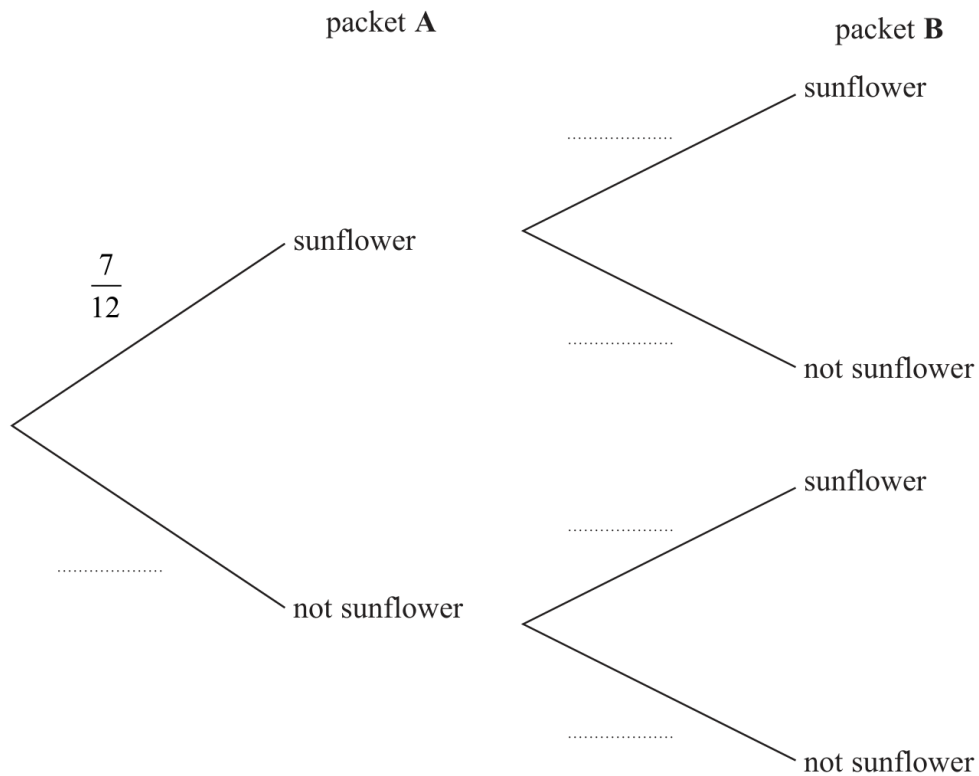
14 Aika has 2 packets of seeds, packet **A** and packet **B**

There are 12 seeds in packet A and 7 of these are sunflower seeds.

There are 15 seeds in packet **B** and 8 of these are sunflower seeds.

Aika is going to take at random a seed from packet **A** and a seed from packet **B**

(a) Complete the probability tree diagram.



(b) Calculate the probability that Aika will take two sunflower seeds.

(2)

Worked solution 13

Chapter: Combined & Conditional Probability

MODULAR UNIT 1

2 marks

Q14(b) – Chapter: Combined & Conditional Probability

Family: Combine independent events • Variant: Find the probability that all independent events occur

Method

$$P(\text{two sunflowers}) = (7/12)(8/15).$$

*Why: Multiply the two independent packet probabilities.***Method**

$$P(\text{two sunflowers}) = 56/180 = 14/45.$$

*Why: Simplify the fraction.***Final answer**probability: $14/45$

18 There are 16 sweets in a bowl.

4 of the sweets are blackcurrant.

5 of the sweets are lemon.

7 of the sweets are orange.

Anna, Ravi and Sam each take at random one sweet from the bowl.

Work out the probability that the 5 lemon sweets are still in the bowl.

Worked solution

January 2020 | Question 18 | 4 marks

Family: Combine selections without replacement • Variant: Use a complement in repeated no-replacement selection

Method / working

1. Non-lemon sweets=11.
2. $P(\text{no lemon in three draws}) = \frac{11}{16} \times \frac{10}{15} \times \frac{9}{14}$.
3. Simplify the product.
4. $P = \frac{33}{112}$.

Final answer: $\frac{33}{112}$

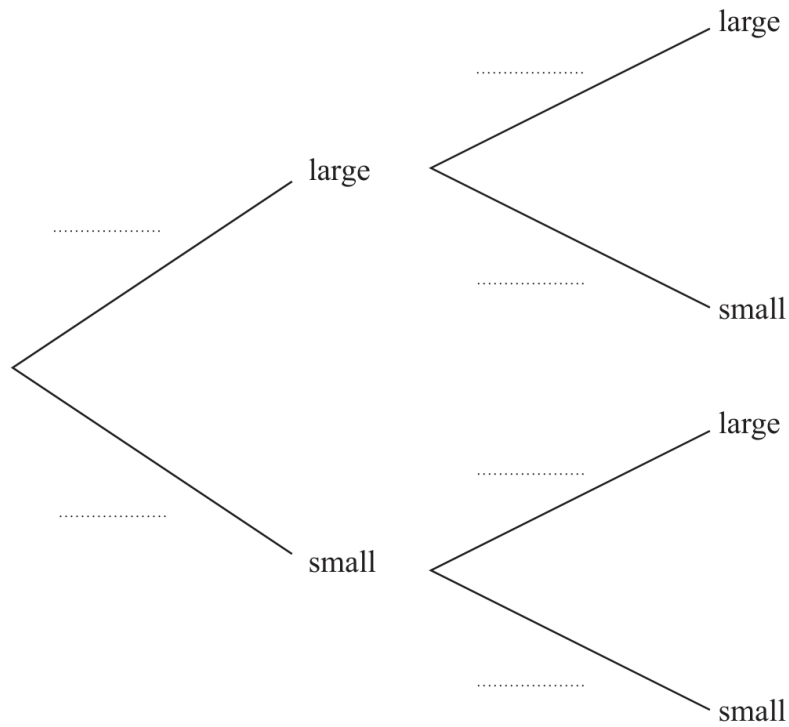
14 There are 20 glasses in a cupboard.

13 of the glasses are large

7 of the glasses are small

Roberto takes at random two glasses from the cupboard.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Roberto takes two small glasses.

(2)

Worked solution

January 2020 | Question 14 | 2 marks

Family: Combine selections without replacement • Variant: Calculate an unordered two-selection event

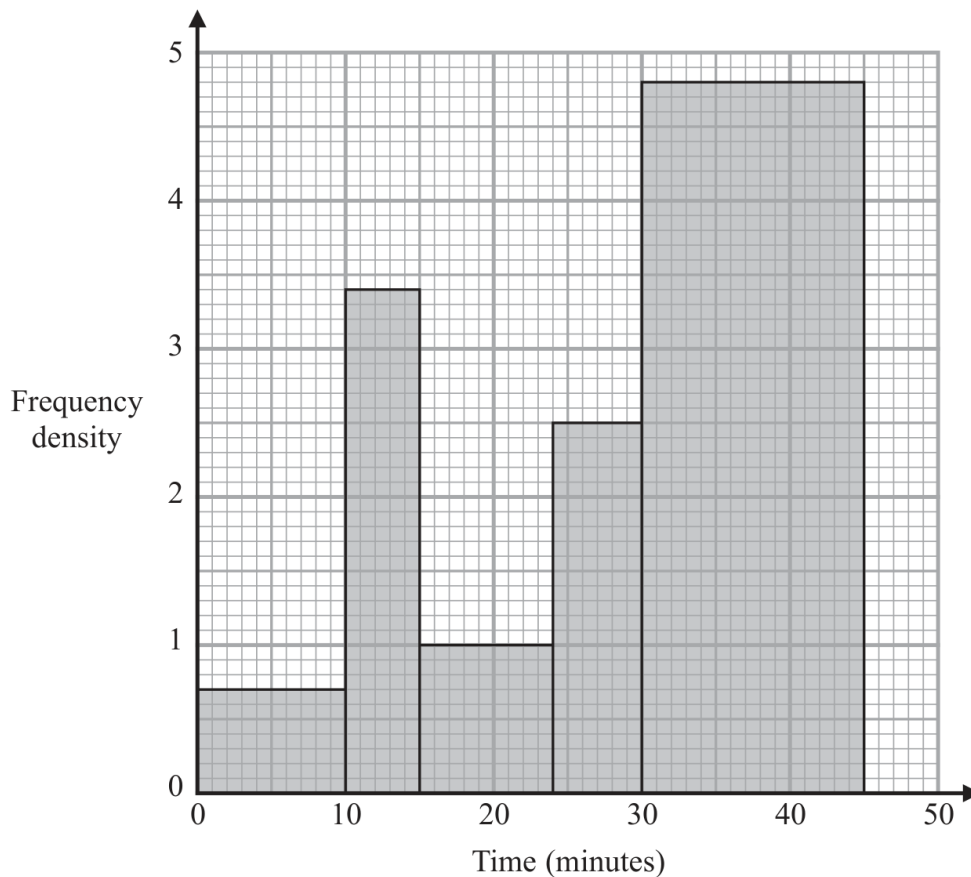
Method / working

1. $P(\text{two small}) = \frac{7}{20} \times \frac{6}{19}$.

2. Simplify to $\frac{21}{190}$.

Final answer: $\frac{21}{190}$

- 18 The histogram gives information about the times, in minutes, that some customers spent in a supermarket.



One of the customers is selected at random.

Given that this customer had spent more than 30 minutes in the supermarket,

- (b) find the probability that this customer spent more than 36 minutes in the supermarket.

(2)

Worked solution

January 2020 | Question 18 | 2 marks

Family: Use conditional probability and restricted sample spaces • Variant: Use a histogram or table as a conditional sample space

Method / working

1. Given more than 30 minutes, use the 30-45 class.
2. The part above 36 is 9 of the 15 minutes, so probability= $\frac{9}{15}=\frac{3}{5}$.

Final answer: $\frac{3}{5}$

(b) Work out the probability that Ding will win exactly one of the games.

(3)

Worked solution

January 2021 | Question 14 | 3 marks

Family: Combine independent events • Variant: Find exactly one, same or different independent outcomes

Method / working

1. Exactly one win: $(0.6 \times 0.1) + (0.4 \times 0.25)$.

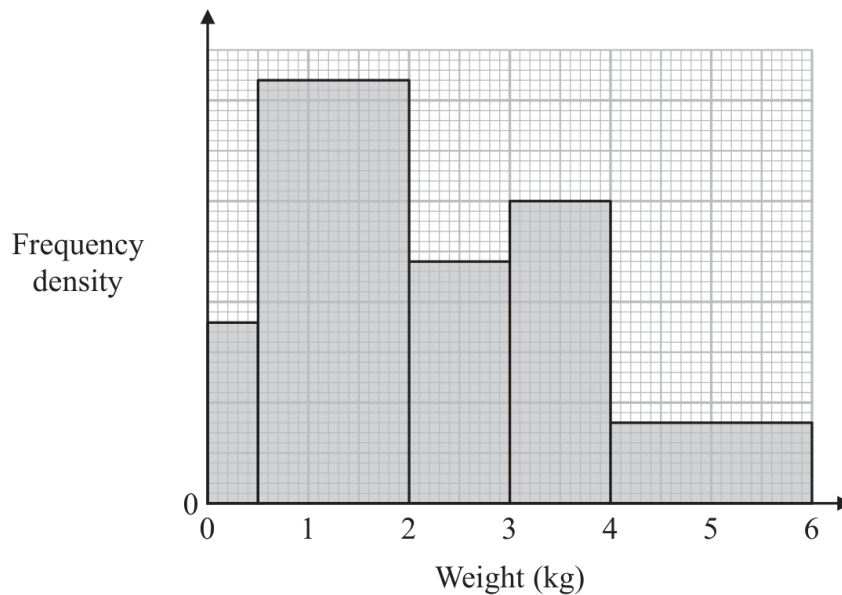
2. This is $0.06 + 0.10 = 0.16$.

3. Therefore the probability is 0.16.

Final answer: 0.16

15 A postman records the weight of each parcel that he delivers.

The histogram shows information about the weights of all the parcels that the postman delivered last Monday. No parcels weighed more than 6 kg.



63 of the parcels that the postman delivered last Monday each had a weight between 0.5 kg and 2 kg.

(a) Work out the total number of parcels the postman delivered last Monday.

(3)

The postman picks at random two of the records of the parcels he delivered last Monday.

(b) Work out an estimate for the probability that each parcel weighed more than 2.25 kg.

(3)

Worked solution

January 2021 | Question 15 | 3 marks

Family: Combine selections without replacement • Variant: Calculate an unordered two-selection event

Method / working

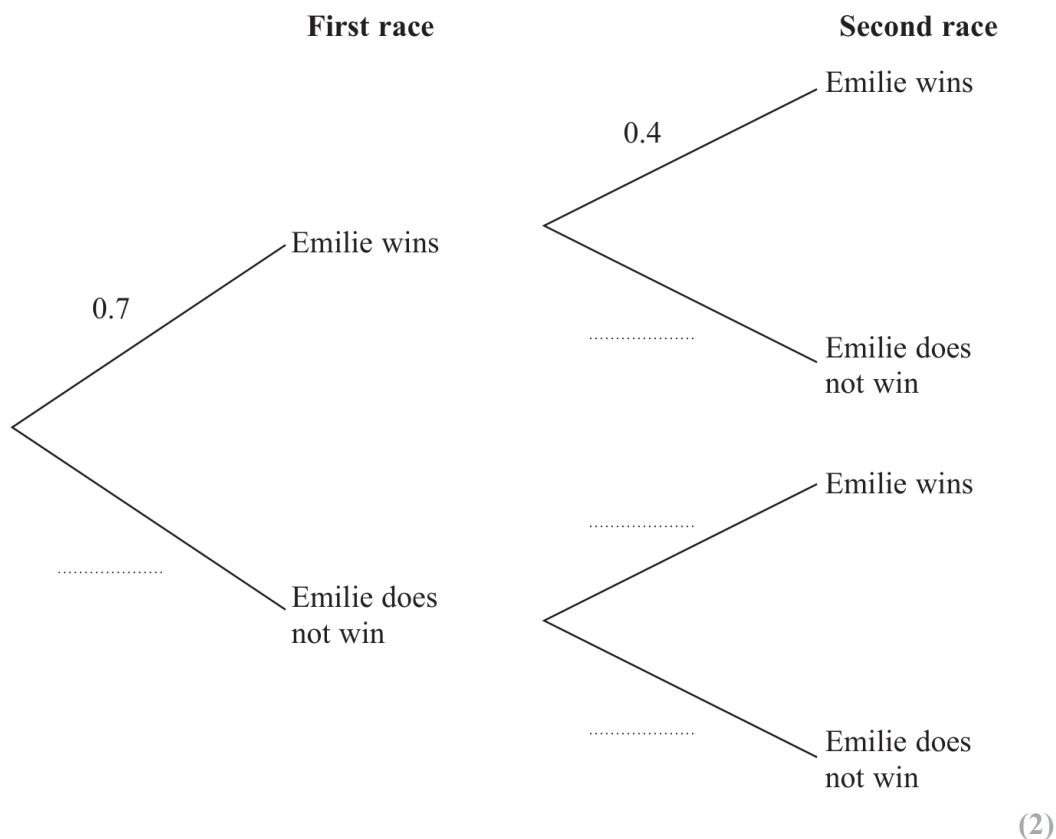
1. Frequency above 2.25 kg is 64 from the histogram estimate.
2. For two independent records, use $(64/142)^2$.
3. This simplifies to $672/3337 \sim 0.201$.

Final answer: $672/3337 \sim 0.201$

13 Emilie takes part in two races.

The probability that she wins the first race is 0.7
 The probability that she wins the second race is 0.4
 The outcomes of the two races are independent.

(a) Complete the probability tree diagram.



(b) Work out the probability that Emilie wins exactly one of the two races.

(3)

Emilie is going to take part in a third race.

If she wins both of the first two races, the probability that she will win the third race is 0.6

If she wins exactly one of the first two races, the probability that she will win the third race is 0.3

(c) Work out the probability that Emilie will win exactly two of the three races.

(3)

Worked solution

June 2021 | Question 13 | 3 marks

Family: Combine independent events • Variant: Find exactly one, same or different independent outcomes

Method / working

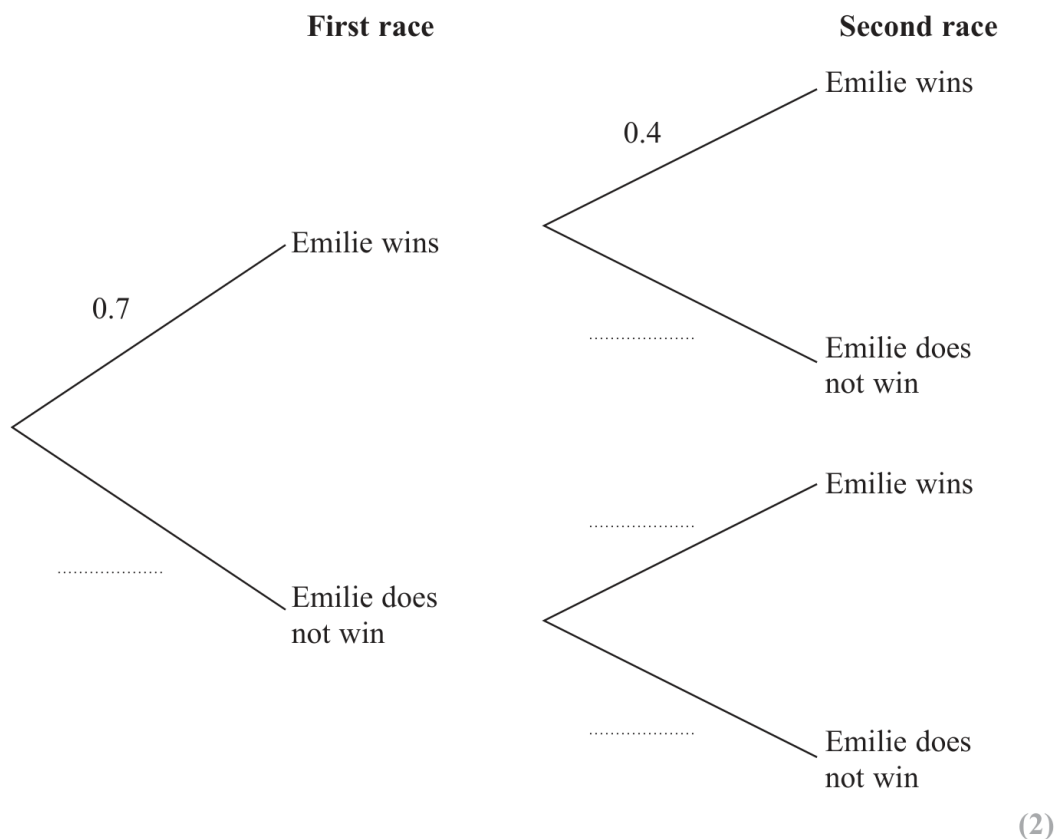
1. Exactly one win has paths $(.7 \times .6)$ and $(.3 \times .4)$.
2. Add them: $.42 + .12 = .54$.
3. State the probability 0.54 (27/50).

Final answer: 0.54

13 Emilie takes part in two races.

The probability that she wins the first race is 0.7
 The probability that she wins the second race is 0.4
 The outcomes of the two races are independent.

(a) Complete the probability tree diagram.



(b) Work out the probability that Emilie wins exactly one of the two races.

(3)

Emilie is going to take part in a third race.

If she wins both of the first two races, the probability that she will win the third race is 0.6

If she wins exactly one of the first two races, the probability that she will win the third race is 0.3

(c) Work out the probability that Emilie will win exactly two of the three races.

(3)

Worked solution

June 2021 | Question 13 | 3 marks

Family: Use conditional probability and restricted sample spaces • Variant: Use branch-dependent conditional probabilities

Method / working

1. If the first two are both wins and the third is not, the probability is $.7 \times .4 \times (1-.6) = .112$.
 2. If exactly one of the first two wins (probability $.54$), then the third win has probability $.3$: $.54 \times .3 = .162$.
 3. Add $.112 + .162$ to obtain $.274$ ($137/500$).
- Final answer: 0.274

23 Pippa has a box containing N pens.

There are only black pens and red pens in the box.

The number of black pens in the box is 3 more than the number of red pens.

Pippa is going to take at random 2 pens from the box.

The probability that she will take a black pen **followed** by a red pen is $\frac{9}{35}$

Find the possible values of N .

Show clear algebraic working.

Worked solution

June 2021 | Question 23 | 5 marks

Family: Combine selections without replacement • Variant: Recover an unknown population from no-replacement probability

Method / working

1. Let red pens= r and black pens= $r+3$, so $N=2r+3$.
2. For black then red, use $\left(\frac{r+3}{2r+3}\right) \cdot \left(\frac{r}{2r+2}\right) = 9/35$.
3. Clear denominators: $35r(r+3) = 9(2r+3)(2r+2)$, which simplifies to $r^2 - 15r + 54 = 0$.
4. Factor: $(r-6)(r-9) = 0$, so $r=6$ or 9 .
5. Then $N=2r+3$ gives $N=15$ or 21 .

Final answer: $N=21$ or 15

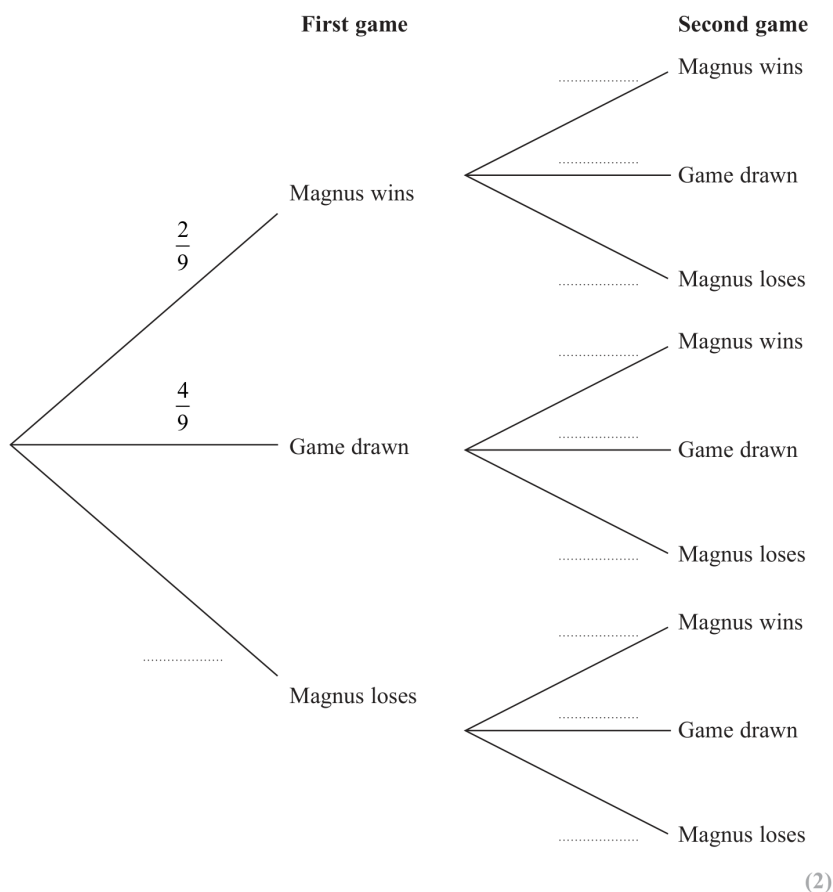
15 Magnus and Garry play 2 games of chess against each other.

The probability that Magnus beats Garry in any game is $\frac{2}{9}$

The probability that any game between Magnus and Garry is drawn is $\frac{4}{9}$

The result of any game is independent of the result of any other game.

(a) Complete the probability tree diagram.



For each game of chess,

the winner gets 2 points and the loser gets 0 points,
when the game is drawn, each player gets 1 point.

(b) Work out the probability that, after 2 games, Magnus and Garry have the same number of points.

(3)

Magnus and Garry now play a third game of chess.

(c) Work out the probability that, after 3 games, Magnus and Garry have the same number of points.

(3)

Worked solution

November 2021 | Question 15 | 3 marks

Family: Combine independent events • Variant: Find exactly one, same or different independent outcomes

Method / working

1. For each chess game:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $P(M \text{ wins}) = \frac{29}{81}$, $\quad P(\text{draw}) = \frac{49}{81}$, $\quad P(M \text{ loses}) = \frac{13}{81}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. After two games, equal points happens if both games are drawn or Magnus wins one and loses one.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. $\left(\frac{49}{81}\right)^2 + 2\left(\frac{29}{81}\right)\left(\frac{13}{81}\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. $=\frac{16}{81} + \frac{12}{81} = \frac{28}{81}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. After three games, equal points means Magnus scores $\frac{1}{3}$ points in total.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. This happens with three draws, or with one win, one draw and one loss.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $\left(\frac{49}{81}\right)^3 + 6\left(\frac{29}{81}\right)\left(\frac{49}{81}\right)\left(\frac{13}{81}\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. $=\frac{64}{729} + \frac{144}{729} = \frac{208}{729}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. Answers: $\left(\frac{28}{81}\right)$, $\left(\frac{208}{729}\right)$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $\frac{28}{81}$, $\frac{208}{729}$.

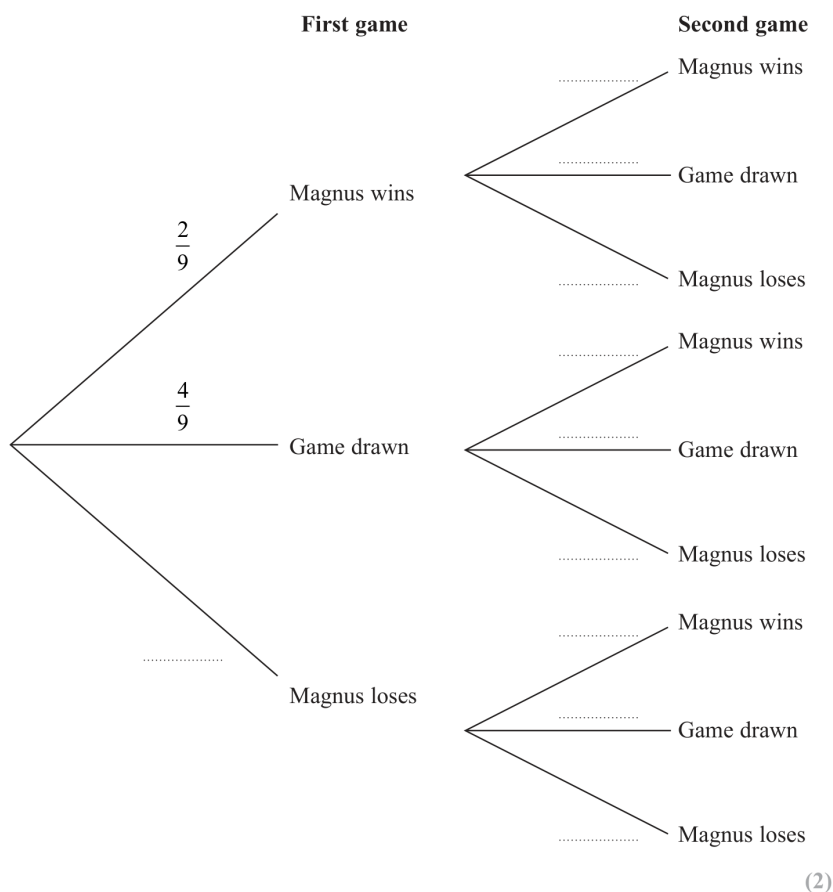
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the winner gets 2 points and the loser gets 0 points,
when the game is drawn, each player gets 1 point.

(b) Work out the probability that, after 2 games, Magnus and Garry have the same number of points.

(3)

Magnus and Garry now play a third game of chess.

(c) Work out the probability that, after 3 games, Magnus and Garry have the same number of points.

(3)

Worked solution

November 2021 | Question 15 | 3 marks

Family: Combine independent events • Variant: Find exactly one, same or different independent outcomes

Method / working

1. For each chess game:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

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7. This happens with three draws, or with one win, one draw and one loss.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $\left(\frac{49}{81}\right)^3 + 6\left(\frac{29}{81}\right)\left(\frac{49}{81}\right)\left(\frac{13}{81}\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. $=\frac{64}{729} + \frac{144}{729} = \frac{208}{729}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

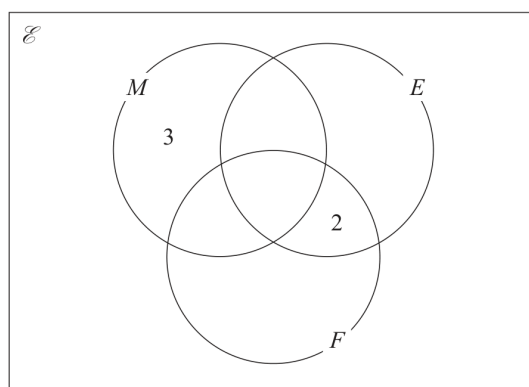
10. Answers: $\left(\frac{28}{81}\right)$, $\left(\frac{208}{729}\right)$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $\frac{28}{81}$, $\frac{208}{729}$.

- 16 There are 32 students in a class.
 In one term these 32 students each took a test in Maths (M), in English (E) and in French (F).
 25 students passed the test in Maths.
 20 students passed the test in English.
 14 students passed the test in French.
 18 students passed the tests in Maths and English.
 11 students passed the tests in Maths and French.
 4 students failed all three tests.
 x students passed all three tests.

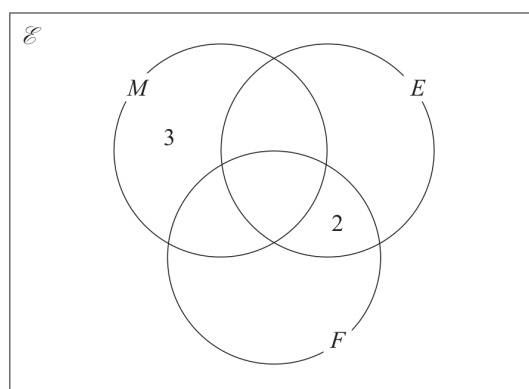
The incomplete Venn diagram gives some more information about the results of the 32 students.



- (a) Use all the given information about the results of students who passed the test in Maths to find the value of x .

$x = \dots\dots\dots$
 (2)

- (b) Use your value of x to complete the Venn diagram to show the number of students in each subset.



(2)

A student who passed the test in Maths is chosen at random.

- (c) Find the probability that this student failed the test in French.

$\dots\dots\dots$
 (1)

Worked solution

November 2021 | Question 16 | 1 marks

Family: Use conditional probability and restricted sample spaces • Variant: Calculate a direct given-that probability

Method / working

1. Use the Maths circle.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $M \text{ only} = 3$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. $M \cap E \text{ only} = 18 - x$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. $M \cap F \text{ only} = 11 - x$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. The total in $\setminus(M)$ is $\setminus(25)$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. $3 + (18 - x) + (11 - x) + x = 25$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. $32 - x = 25$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $x = 7$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. So:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. $M \cap E \text{ only} = 11$, $\quad M \cap F \text{ only} = 4$, $\quad E \cap F \text{ only} = 2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

11. $E \text{ only} = 20 - 11 - 2 - 7 = 0$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

12. $F \text{ only} = 14 - 4 - 2 - 7 = 1$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

13. For a student who passed Maths, failed French means $\setminus(M)$ but not $\setminus(F)$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

14. $\frac{3+11}{25} = \frac{14}{25}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

15. Answers: $\setminus(x=7)$, and $\setminus(\frac{14}{25})$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $x=7$, and $\frac{14}{25}$.

16 A box contains 15 counters.

There are 4 red counters, 5 green counters and the rest are yellow counters.

Niklas takes at random a counter from the box and writes down the colour of his counter.
He then puts the counter back into the box.

Sasha then takes at random a counter from the box and writes down the colour of her counter.

Work out the probability that the counters taken by Niklas and Sasha both have the same colour.

Worked solution

November 2021 | Question 16 | 3 marks

Family: Combine set counts with sequential probability • Variant: Compare with-replacement and without-replacement models

Method / working

1. There are $\binom{4}{1}$ red, $\binom{5}{1}$ green and $\binom{6}{1}$ yellow counters.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. The first counter is replaced, so the probabilities are the same for Niklas and Sasha.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. $P(\text{same colour}) =$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. $\left(\frac{4}{15}\right)^2 +$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. $\left(\frac{5}{15}\right)^2 +$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. $\left(\frac{6}{15}\right)^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. $= \frac{16+25+36}{225}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $= \frac{77}{225}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. Answer: $\left(\frac{77}{225}\right)$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $\frac{77}{225}$.

20 A bag contains X counters.

There are only red counters and blue counters in the bag.

There are 4 more blue counters than red counters in the bag.

Finty takes at random 2 counters from the bag.

The probability that Finty takes 2 blue counters from the bag is $\frac{3}{8}$

Work out the value of X .

Show clear algebraic working.

(Total for Question 20 is 5 marks)

Answer reference | May-Nov 2020 | Question 20

Family: Combine selections without replacement • Variant: Recover an unknown population from no-replacement probability

20	$\left(\frac{X+4}{2}\right) \frac{X}{X} (= \frac{X+4}{2X}) \text{ or } \left(\frac{X+4}{2}\right) - 1 (= \frac{X+2}{2X-2})$	eg, where b = number of blue counters $\frac{b}{2b-4} \text{ or } \frac{b-1}{2b-5}$	eg, where r = number of red counters $\frac{r+4}{2r+4} \text{ or } \frac{r+3}{2r+3}$		M1	for making a correct start by finding the probability of the first counter being blue for their method
	eg $\frac{X+4}{2X} \times \frac{X+2}{2X-2}$	eg $\frac{b}{2b-4} \times \frac{b-1}{2b-5}$	eg $\frac{r+4}{2r+4} \times \frac{r+3}{2r+3}$		M1	oe correct calculation for 2 blue (using one variable)
	eg $8(X^2 + 6X + 8) = 3(4X^2 - 4X)$	eg $8b(b-1) = 3(2b-4)(2b-5)$	eg $8(r+4)(r+3) = 3(2r+4)(2r+3)$		M1	dep for a correct equation with no algebraic fractions eg could have $X^2 + 6X + 8 = \frac{3}{8}(4X^2 - 4X)$
	Eg $4X^2 - 60X - 64 (= 0)$ or $X^2 - 15X - 16 (= 0)$ oe	eg $4b^2 - 46b + 60 (= 0)$ or $2b^2 - 23b + 30 (= 0)$ oe	eg $4r^2 - 14r - 60 (= 0)$ or $2r^2 - 7r - 30 (= 0)$ oe		M1	for rearranging their equation to a correct 3 term quadratic
				16	5	A1 cao dep on M4
Total 5 marks						

Worked solution

May-Nov 2020 | Question 20 | 5 marks

Family: Combine selections without replacement • Variant: Recover an unknown population from no-replacement probability

Official final mark-scheme pages are embedded immediately after this question.

22 There are 15 buttons in a box.

3 of the buttons are red
2 of the buttons are pink
10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

.....
(Total for Question 22 is 3 marks)

Worked solution

November 2025 | Question 22 | 3 marks

Family: Combine selections without replacement • Variant: Calculate one ordered no-replacement path

Method

There are 2 pink buttons.

At least one pink button is still in the box unless both pink buttons are taken.

So use the complement.

The probability that both pink buttons are taken is

$$\frac{\binom{2}{2}\binom{13}{1}}{\binom{15}{3}}$$

$$= \frac{13}{455}$$

$$= \frac{1}{35}$$

Therefore,

P(at least one pink is still in the box)

$$= 1 - \frac{1}{35}$$

$$= \frac{34}{35}$$

Answer: $\frac{34}{35}$